

برای آنالیز از زبان پایتون استفاده شده. از آنجایی که تعداد افراد سالم (GROUP1) کم بود از تکنیک GAN (به فارسی شبکه مولد متخاصم) که یکی از تکنیک های تولید داده بسیار شبیه به داده های واقعی است استفاده شده. (لینک مقاله GAN: [Generative Adversarial Networks](#) [1406.2661])

برای ساختن شبکه GAN از فریمورک PyTorch (یکی از فریمورک های بسیار مطرح در زمینه هوش مصنوعی، منتشر شده توسط متا استفاده شده لینک مقاله : [PyTorch: An Imperative Style, High-Performance Deep Learning Library](#) [1912.01703])

برای رسم نمودارها از کتابخانه Matplotlib استفاده شده است.

برای خواندن دیتاست ها و تولید دیتاست های csv در اکسل از کتابخانه pandas استفاده شده.

برای محاسبه میانگین، انحراف معیار و دیگر محاسبات عددی از کتابخانه numpy استفاده شده.

از الگوریتم K-Means Clustering برای دسته بندی اتوماتیک (بدون نظارت به انگلیسی Unsupervised) برای دسته بندی گروه ۱ و ۲ استفاده شده است تا با اندازه گیری همزمان پارامترهای z ، voxel ، BSC و cop ببینیم آیا تفاوت معناداری بین افراد سالم و مریض در MRI آنها وجود دارد یا خیر لینک مقاله: [\[2304.09989\] CKmeans and FCKmeans : Two deterministic initialization procedures for Kmeans algorithm using a modified crowding distance](#)

برای پیاده سازی الگوریتم KMeans Clustering از کتابخانه sklearn استفاده شده است.

لینک کدهای تولید دیتا با الگوریتم GAN به زبان پایتون:

<https://github.com/ArminKardan/jal/blob/master/gan.py>

لینک کدهای KMeans Clustering:

<https://github.com/ArminKardan/jal/blob/master/kmeans.py>

برای ارزیابی نرمالیتی دیتا از روش Shapiro-Wilk استفاده شده اما به دلیل کمبود دیتاها، دیتا های پرت میتواند نتیجه این الگوریتم را تغییر دهد به همین منظور برای اثبات نرمالیتی دیتا از روش Q-Q استفاده شده است و برای پیاده سازی Q-Q و Shapiro از کتابخانه scipy استفاده شده است.

لینک کدهای Q-Q برای نمایش نرمالیتی:

<https://github.com/ArminKardan/jal/blob/master/QQ.py>

برای بررسی و رسم نمودارهای t-test از کتابخانه scipy استفاده شده است.

لینک کدهای t-test:

<https://github.com/ArminKardan/jal/blob/master/diagram.py>

لینک های دیتاست ها:

گروه ۱ استم چپ (آگمنت شده):

<https://github.com/ArminKardan/jal/blob/master/G1LT-A.csv>

گروه ۱ استم چپ (اورجینال):

<https://github.com/ArminKardan/jal/blob/master/G1LT.csv>

گروه ۲ استم چپ:

<https://github.com/ArminKardan/jal/blob/master/G2LT.csv>

گروه ۱ استم راست (آگمنت شده):

<https://github.com/ArminKardan/jal/blob/master/G1RT-A.csv>

گروه ۱ استم راست (اورجینال):

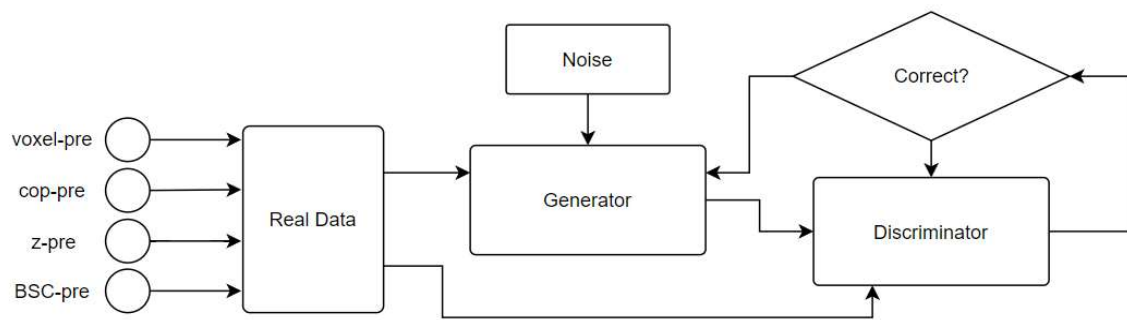
<https://github.com/ArminKardan/jal/blob/master/G1RT.csv>

گروه ۲ استم راست:

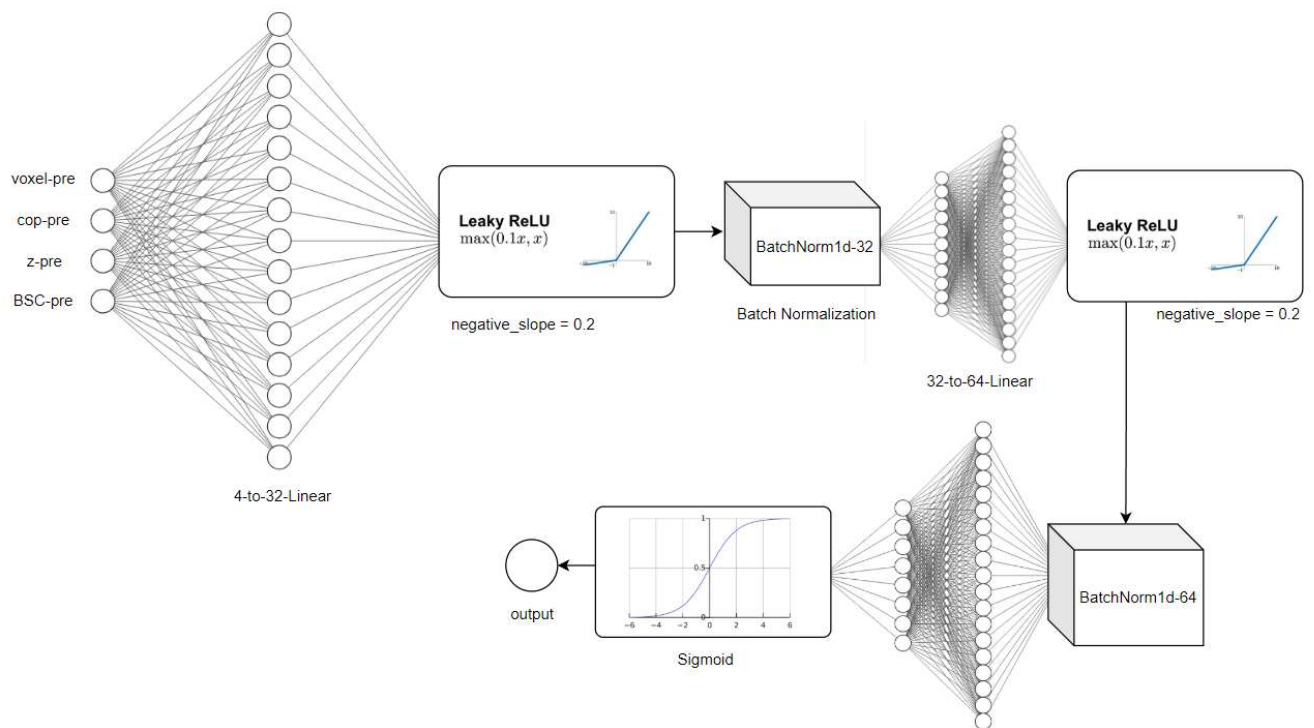
<https://github.com/ArminKardan/jal/blob/master/G2RT.csv>

شاخص ARI:

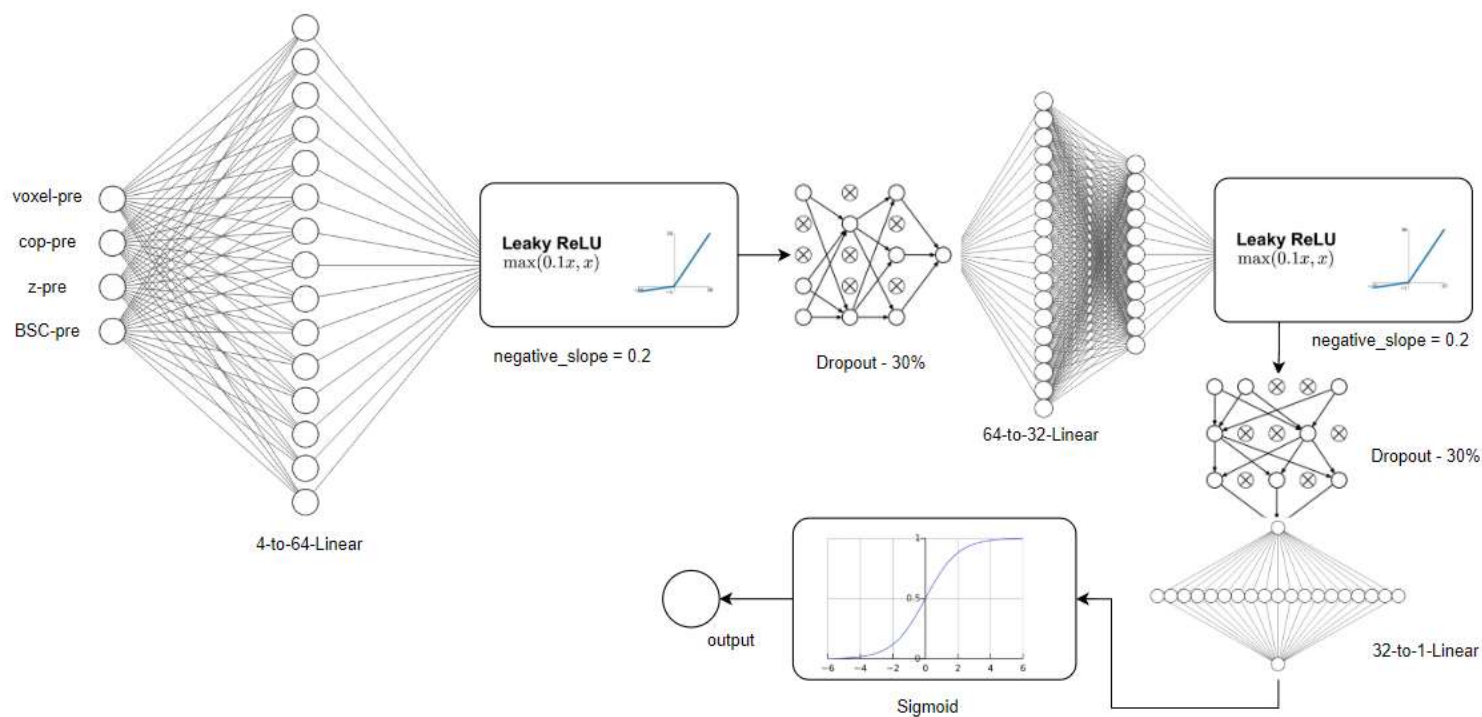
[Understanding the Adjusted Rand Index and Other Partition Comparison Indices Based on Counting Object Pairs | Journal of Classification](#)



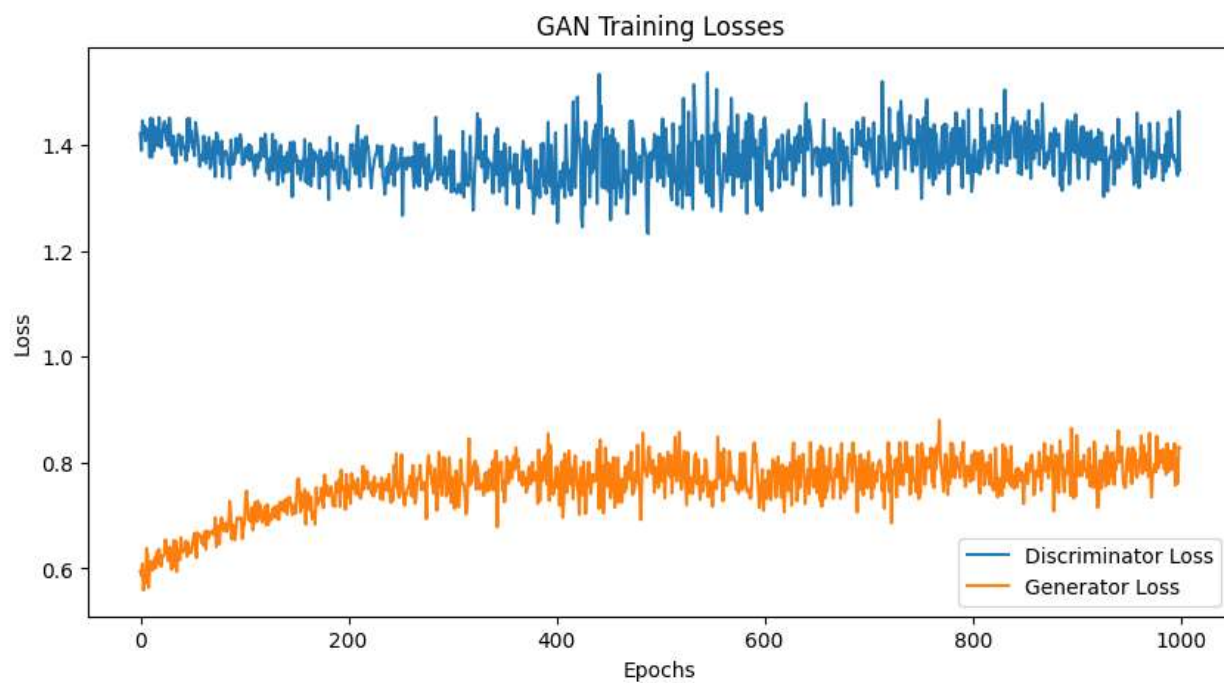
GAN DIAGRAM



GAN-Generator Unit Diagram



GAN-Discriminator Unit Diagram



RIGHT STEM - GROUP1 – Augmented by GAN Network data result:

coppre voxelpre zpre BSCpre

0	189.10	65.0	7.086	0.200
1	193.69	61.0	3.747	0.200
2	181.00	57.0	4.099	0.115
3	116.30	55.0	4.750	0.160
4	126.70	76.0	4.950	0.200
5	173.60	53.0	4.330	0.250
6	126.80	64.0	3.810	0.160
7	75.39	73.0	3.761	0.180
8	154.23	62.0	3.660	0.240
9	108.00	74.0	4.507	0.150
10	181.10	46.0	2.830	0.196
11	150.54	61.0	4.092	0.176
12	138.75	57.0	2.766	0.222
13	97.11	78.0	4.168	0.216
14	160.04	78.0	6.349	0.173
15	100.88	60.0	5.005	0.208
16	157.72	72.0	4.357	0.200
17	156.99	66.0	5.747	0.191
18	126.73	77.0	3.730	0.154
19	156.02	72.0	3.785	0.153
20	112.36	72.0	2.469	0.162
21	202.70	61.0	3.706	0.155
22	113.60	64.0	5.498	0.216
23	137.71	71.0	3.423	0.176
24	153.21	46.0	4.094	0.203
25	103.95	50.0	4.998	0.192
26	141.47	54.0	4.421	0.186
27	198.31	73.0	3.512	0.183
28	84.53	68.0	5.175	0.199
29	114.09	68.0	3.291	0.247
30	166.40	63.0	4.520	0.122
31	198.41	72.0	4.552	0.181
32	173.60	58.0	4.487	0.173
33	129.98	72.0	4.585	0.125
34	140.47	71.0	4.315	0.205
35	172.74	78.0	3.189	0.171
36	177.98	77.0	3.629	0.191
37	148.16	60.0	4.633	0.134
38	148.71	75.0	3.716	0.145
39	154.01	64.0	4.335	0.123

RIGHT STEM - GROUP2:

0	189.10	65.0	7.086	0.200
1	193.69	61.0	3.747	0.200
2	181.00	57.0	4.099	0.115
3	116.30	55.0	4.750	0.160
4	126.70	76.0	4.950	0.200
5	173.60	53.0	4.330	0.250
6	126.80	64.0	3.810	0.160
7	75.39	73.0	3.761	0.180
8	154.23	62.0	3.660	0.240
9	108.00	74.0	4.507	0.150
10	134.50	57.0	4.399	0.123
11	171.49	59.0	4.073	0.202
12	105.21	72.0	3.930	0.123
13	192.69	54.0	3.844	0.249
14	79.84	76.0	4.008	0.142
15	169.38	55.0	4.084	0.177
16	183.00	53.0	4.067	0.246
17	181.06	57.0	3.896	0.221
18	99.19	74.0	4.171	0.115
19	162.58	63.0	4.055	0.184
20	145.33	65.0	3.842	0.216
21	100.69	64.0	3.927	0.122
22	121.42	76.0	4.395	0.152
23	182.40	56.0	4.030	0.226
24	137.75	63.0	4.169	0.116
25	143.48	76.0	3.818	0.232
26	162.24	63.0	3.828	0.234
27	193.52	53.0	3.766	0.250
28	178.98	53.0	3.831	0.248
29	83.16	75.0	4.134	0.127
30	112.76	69.0	4.132	0.157
31	149.23	76.0	4.012	0.151
32	175.44	58.0	3.851	0.239
33	183.81	58.0	3.799	0.246
34	124.71	62.0	3.784	0.150
35	143.41	70.0	3.837	0.173
36	148.94	67.0	3.838	0.187
37	113.69	72.0	3.923	0.180
38	92.75	76.0	4.006	0.118
39	98.53	75.0	4.226	0.117

RIGHT STEM - GROUP1-Augmented vs GROUP2 T-Test Report:

T-test for coppre:

t-statistic: 0.414

p-value: 0.679

T-test for voxelpre:

t-statistic: 0.480

p-value: 0.632

T-test for zpre:

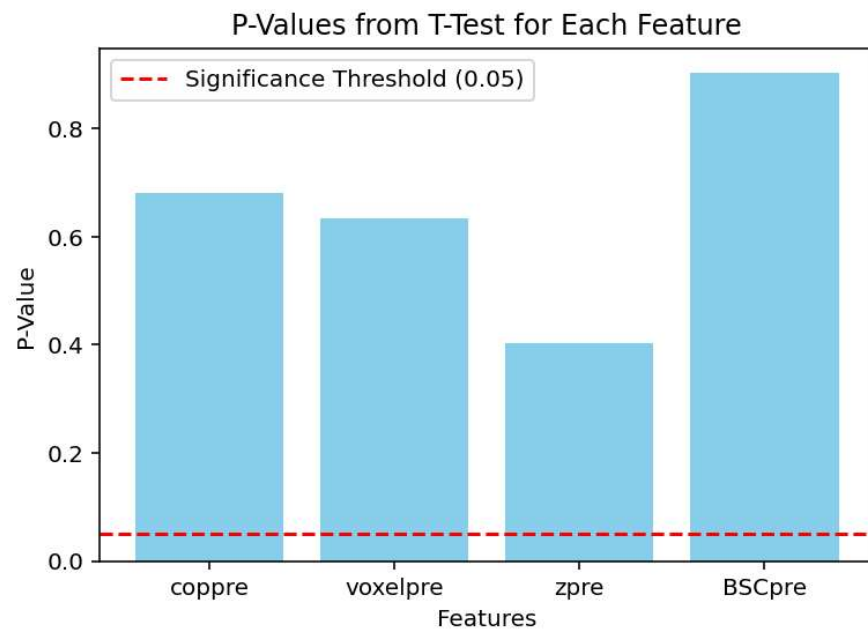
t-statistic: 0.840

p-value: 0.402

T-test for BSCpre:

t-statistic: -0.123

p-value: 0.902



RIGHT STEM - GROUP1 (Original) vs GROUP2 T-Test Result:

T-test for coppre:

t-statistic: 0.122

p-value: 0.903

T-test for voxelpre:

t-statistic: -0.232

p-value: 0.817

T-test for zpre:

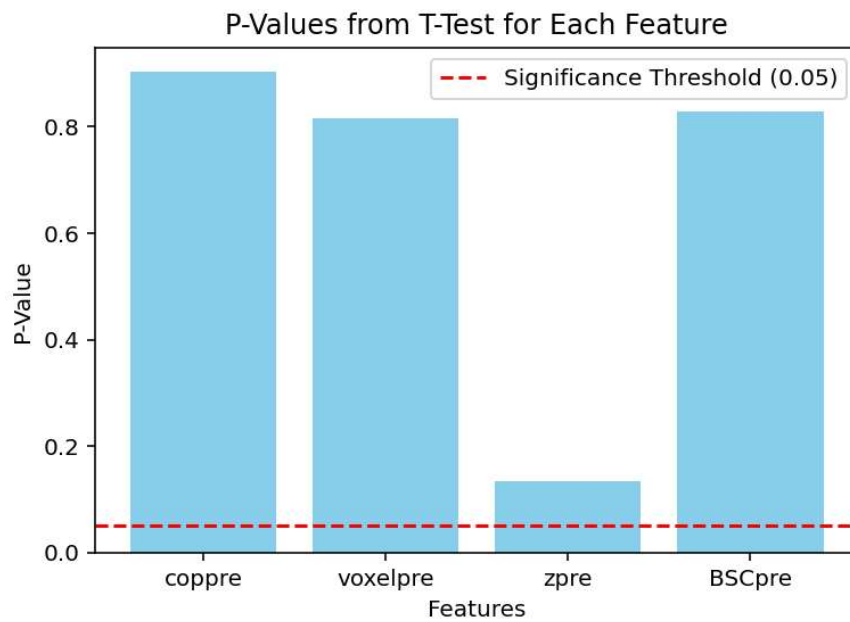
t-statistic: 1.525

p-value: 0.133

T-test for BSCpre:

t-statistic: 0.217

p-value: 0.828



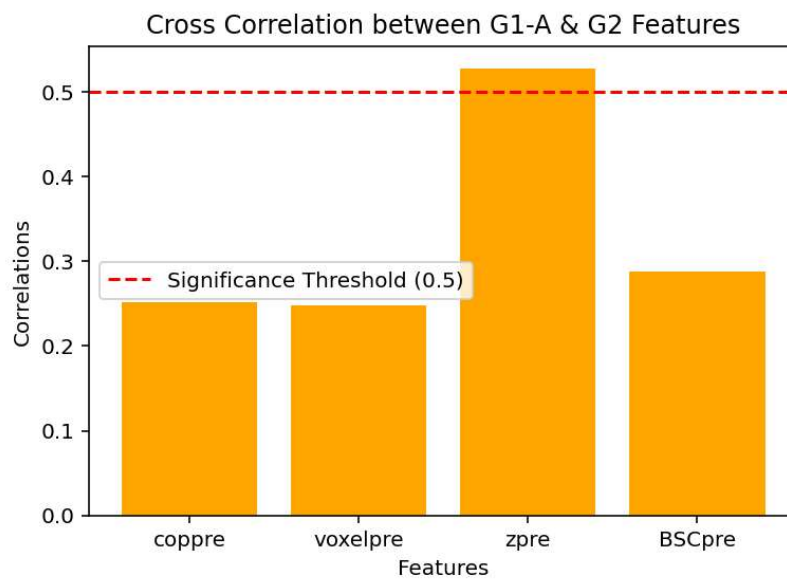
RIGHT STEM - Correlation [GROUP1-Augmented vs GROUP2] :

Correlation coppre: 0.25

Correlation voxelpre: 0.25

Correlation zpre: 0.53

Correlation BSCpre: 0.29



RIGHT STEM - GROUP1-Augmented Statistics:

Mean coppre G1-A: 146.08

Mean voxelpre G1-A: 65.6

Mean zpre G1-A: 4.25

Mean BSCpre G1-A: 0.18

STDEV coppre: 32.31

STDEV voxelpre: 8.87

STDEV zpre: 0.91

STDEV BSCpre: 0.03

RIGHT STEM - GROUP1 Statistics:

Mean coppre: 144.48

Mean voxelpre: 64.0

Mean zpre: 4.47

Mean BSCpre: 0.19

STDEV coppre: 37.7

STDEV voxelpre: 7.68

STDEV zpre: 0.97

STDEV BSCpre: 0.04

RIGHT STEM - GROUP2 Statistics:

Mean coppre: 142.9

Mean voxelpre: 64.68

Mean zpre: 4.11

Mean BSCpre: 0.18

STDEV coppre: 35.39

STDEV voxelpre: 8.13

STDEV zpre: 0.55

STDEV BSCpre: 0.05

LEFT STEM - GROUP1 – Augmented by GAN Network data result:

	copp	voxelp	zpre	BSCpre
0	119.000	60.000	5.438	0.120
1	159.110	60.000	4.600	0.190
2	181.000	51.000	5.023	0.125
3	212.800	58.000	4.208	0.100
4	133.200	70.000	4.355	0.200
5	156.240	48.000	5.710	0.150
6	222.900	58.000	4.900	0.160
7	147.900	65.000	4.780	0.150
8	200.300	60.000	4.420	0.100
9	208.000	72.000	6.510	0.200
10	116.029	52.875	4.507	0.090
11	142.854	56.683	6.070	0.133
12	115.388	56.332	5.947	0.143
13	157.858	84.754	5.598	0.141
14	188.978	48.209	6.110	0.155
15	172.311	55.817	5.903	0.174
16	172.629	69.254	5.714	0.167
17	152.307	51.494	6.334	0.164
18	152.217	69.109	4.614	0.152
19	216.684	69.446	4.757	0.169
20	142.363	52.023	4.932	0.130
21	134.150	59.185	4.956	0.111
22	175.693	63.070	5.201	0.176
23	194.514	51.652	4.429	0.193
24	150.879	58.139	5.614	0.128
25	136.992	64.006	5.249	0.124
26	159.289	57.999	4.918	0.183
27	204.219	62.571	5.042	0.097
28	199.806	66.114	5.176	0.105
29	122.133	70.493	3.581	0.182
30	214.894	49.191	4.792	0.148
31	186.978	59.783	4.228	0.139
32	206.946	60.868	4.489	0.138
33	196.887	66.322	5.308	0.132
34	186.113	66.366	4.293	0.158
35	178.298	53.134	4.366	0.106
36	142.368	51.148	5.644	0.161
37	65.987	49.870	4.690	0.126
38	191.583	57.693	4.396	0.144

39 154.613 61.407 5.079 0.136

LEFT STEM – GROUP2:

coppre voxelpre zpre BSCpre

0	168.20	53.0	7.7180	0.150
1	44.55	56.0	0.6312	0.025
2	148.20	64.0	2.5460	0.150
3	130.40	53.0	5.1000	0.102
4	68.58	50.0	3.0670	0.150
5	359.50	53.0	4.3490	0.105
6	171.70	62.0	5.5680	0.010
7	96.86	59.0	4.5140	0.100
8	201.90	55.0	3.8100	0.020
9	71.07	57.0	3.5360	0.050
10	66.27	65.0	3.2700	0.050
11	78.24	55.0	3.6810	0.250
12	105.40	55.0	3.7600	0.037
13	75.90	64.0	3.0810	0.180
14	250.50	55.0	6.4960	0.200
15	108.20	58.0	5.0340	0.170
16	50.10	61.0	3.1920	0.050
17	122.70	53.0	2.1080	0.100
18	43.48	60.0	1.8810	0.070
19	226.30	60.0	6.7410	0.200
20	207.10	48.0	4.7650	0.120
21	169.90	53.0	7.7910	0.175
22	59.85	59.0	3.3200	0.050
23	221.10	58.0	3.5130	0.090
24	89.19	55.0	2.5380	0.070
25	141.30	56.0	3.6690	0.070
26	105.70	50.0	5.0020	0.200
27	63.03	59.0	3.4150	0.200
28	125.00	64.0	3.5630	0.100
29	168.60	58.0	5.6170	0.200
30	90.83	61.0	3.3960	0.100
31	38.08	58.0	1.7570	0.055
32	54.89	65.0	2.8000	0.055
33	63.89	51.0	4.0260	0.060
34	111.20	61.0	4.9200	0.110
35	52.96	51.0	2.2890	0.050
36	104.50	50.0	5.4430	0.090
37	197.80	59.0	1.8560	0.075

38 111.70 60.0 4.1080 0.060

LEFT STEM - GROUP1-Augmented vs GROUP2 T-Test Report:

T-test for coppre:

t-statistic: 3.62

p-value: 0.00052

T-test for voxelpre:

t-statistic: 1.990

p-value: 0.0501

T-test for zpre:

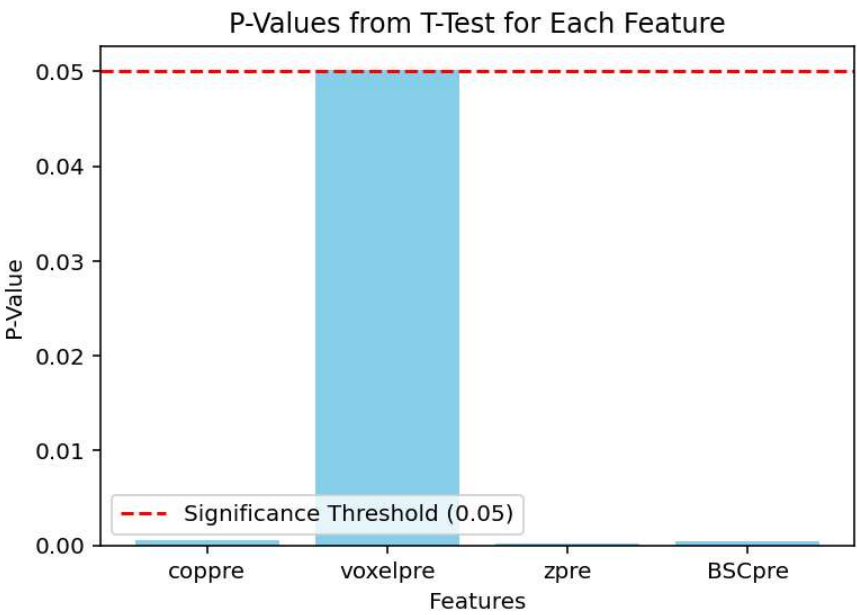
t-statistic: 4.02

p-value: 0.000133

T-test for BSCpre:

t-statistic: 3.69

p-value: 0.000413



LEFT STEM - GROUP1 (Original) vs GROUP2 T-Test Result:

T-test for coppre:

t-statistic: 2.267

p-value: 0.0280

T-test for voxelpre:

t-statistic: 1.703

p-value: 0.095

T-test for zpre:

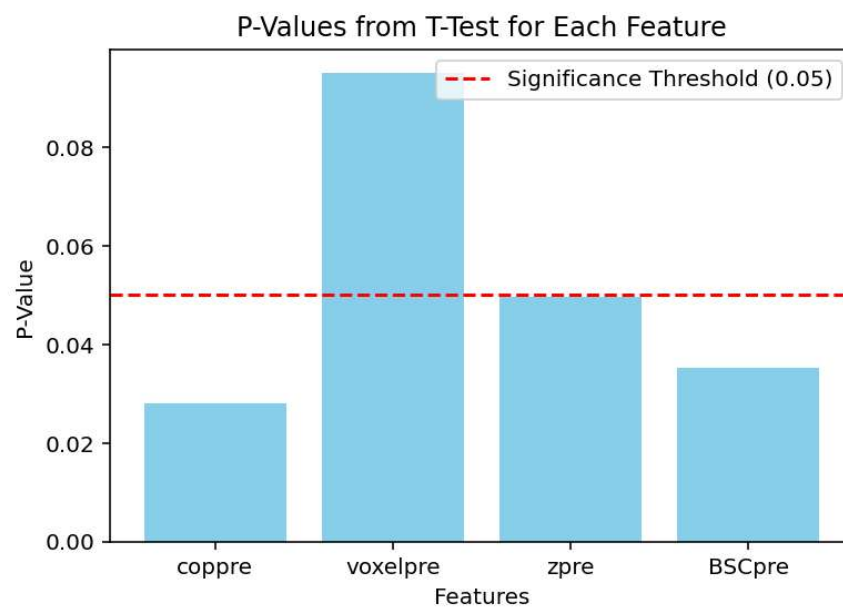
t-statistic: 2.015

p-value: 0.0496

T-test for BSCpre:

t-statistic: 2.166

p-value: 0.035



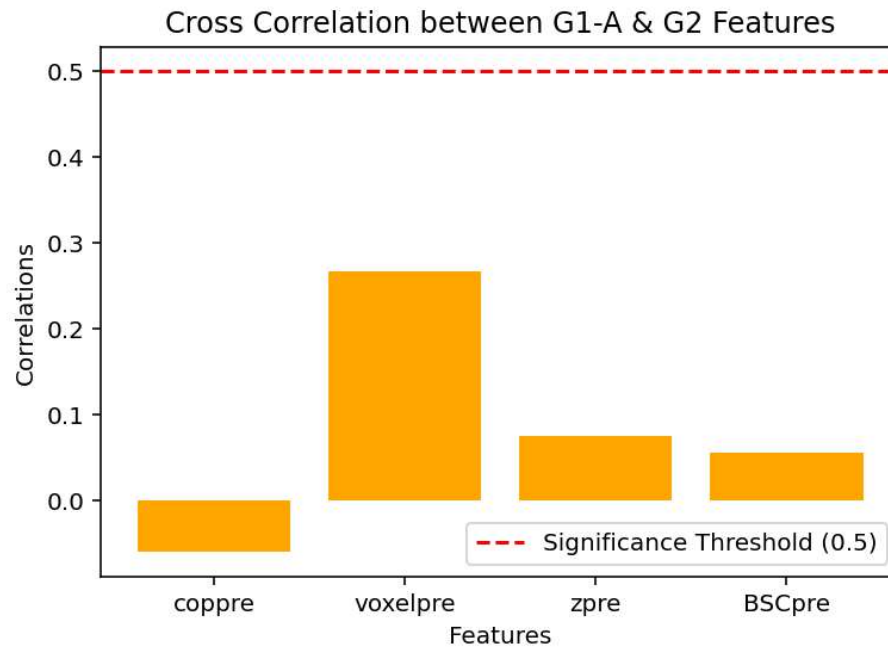
LEFT STEM - Correlation [GROUP1-Augmented vs GROUP2] :

Correlation coppre: -0.059

Correlation voxelpre: 0.2669

Correlation zpre: 0.074

Correlation BSCpre: 0.055



LEFT STEM - GROUP1-Augmented Statistics:

Mean coppre G1-A: 167.12

Mean voxelpre G1-A: 59.89

Mean zpre G1-A: 5.05

Mean BSCpre G1-A: 0.15

STDEV coppre G1-A: 34.67

STDEV voxelpre G1-A: 7.89

STDEV zpre G1-A: 0.67

STDEV BSCpre G1-A: 0.03

LEFT STEM - GROUP1 Statistics:

Mean coppre G1: 174.04

Mean voxelpre G1: 60.2

Mean zpre G1: 4.99

Mean BSCpre G1: 0.15

STDEV coppre G1: 34.21

STDEV voxelpre G1: 7.08

STDEV zpre G1: 0.68

STDEV BSCpre G1: 0.04

LEFT STEM - GROUP2 Statistics:

Mean coppre G2: 122.17

Mean voxelpre G2: 57.03

Mean zpre G2: 3.95

Mean BSCpre G2: 0.11

STDEV coppre G2: 68.71

STDEV voxelpre G2: 4.52

STDEV zpre G2: 1.58

STDEV BSCpre G2: 0.06

Normality test with Shapiro-Wilk Test:

RIGHT STEM GROUP1 - coppre:

Shapiro-Wilk test statistic: 0.9347502863043512

P-value: 0.4962041637397511

Data looks like a normal distribution

RIGHT STEM GROUP1 - voxelpre:

Shapiro-Wilk test statistic: 0.9317832509786342

P-value: 0.4656600163517724

Data looks like a normal distribution

RIGHT STEM GROUP1 - zpre:

Shapiro-Wilk test statistic: 0.7550868371782665

P-value: 0.004115879333578123

Data does not look like a normal distribution

RIGHT STEM GROUP1 - BSCpre:

Shapiro-Wilk test statistic: 0.9609977486590815

P-value: 0.797187667496022

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - coppre:

Shapiro-Wilk test statistic: 0.9787502959937275

P-value: 0.6432029747873634

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - voxelpre:

Shapiro-Wilk test statistic: 0.9468120674994568

P-value: 0.05893671039559701

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - zpre:

Shapiro-Wilk test statistic: 0.9547641239869776

P-value: 0.11067094925897186

Data looks like a normal distribution

RIGHT STEM GROUP1 Augmented - BSCpre:

Shapiro-Wilk test statistic: 0.9790087708317297

P-value: 0.6527620151038658

Data looks like a normal distribution

RIGHT STEM GROUP2 - coppre:

Shapiro-Wilk test statistic: 0.9414297035763521

P-value: 0.03864580228908557

Data does not look like a normal distribution

RIGHT STEM GROUP2 - voxelpre:

Shapiro-Wilk test statistic: 0.9000235257065232

P-value: 0.0019147291238539451

Data does not look like a normal distribution

RIGHT STEM GROUP2 - zpre:

Shapiro-Wilk test statistic: 0.5585113513952625

P-value: 8.860769496060262e-10

Data does not look like a normal distribution

RIGHT STEM GROUP2 - BSCpre:

Shapiro-Wilk test statistic: 0.9134757493092074

P-value: 0.004830024870056205

Data does not look like a normal distribution

LEFT STEM GROUP1 - coppre:

Shapiro-Wilk test statistic: 0.9415758512786505

P-value: 0.5706781090100408

Data looks like a normal distribution

LEFT STEM GROUP1 - voxelpre:

Shapiro-Wilk test statistic: 0.9444611265299422

P-value: 0.6036315202360633

Data looks like a normal distribution

LEFT STEM GROUP1 - zpre:

Shapiro-Wilk test statistic: 0.9110216629160103

P-value: 0.2880827649566656

Data looks like a normal distribution

LEFT STEM GROUP1 - BSCpre:

Shapiro-Wilk test statistic: 0.9079759242863261

P-value: 0.26737515719357663

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - coppre:

Shapiro-Wilk test statistic: 0.9615609841992756

P-value: 0.20099180597729982

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - voxelpre:

Shapiro-Wilk test statistic: 0.9487451710201183

P-value: 0.07455498596278698

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - zpre:

Shapiro-Wilk test statistic: 0.9710103138586211

P-value: 0.4031273395836985

Data looks like a normal distribution

LEFT STEM GROUP1 Augmented - BSCpre:

Shapiro-Wilk test statistic: 0.9776450325797402

P-value: 0.6182274923221711

Data looks like a normal distribution

LEFT STEM GROUP2 - coppre:

Shapiro-Wilk test statistic: 0.8956698870973706

P-value: 0.001661701624117287

Data does not look like a normal distribution

LEFT STEM GROUP2 - voxelpre:

Shapiro-Wilk test statistic: 0.9680980893792391

P-value: 0.32765361461131115

Data looks like a normal distribution

LEFT STEM GROUP2 - zpre:

Shapiro-Wilk test statistic: 0.9670293946306404

P-value: 0.30305369562342976

Data looks like a normal distribution

LEFT STEM GROUP2 - BSCpre:

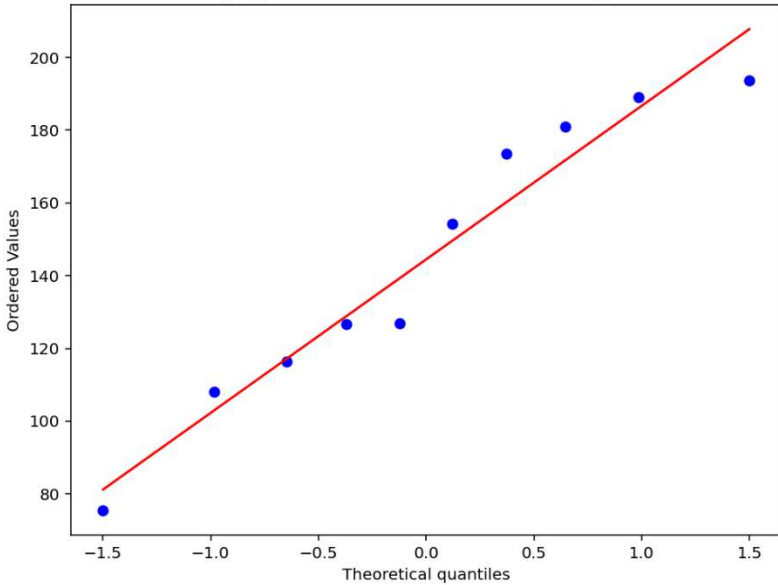
Shapiro-Wilk test statistic: 0.9293010855318422

P-value: 0.017067319779668567

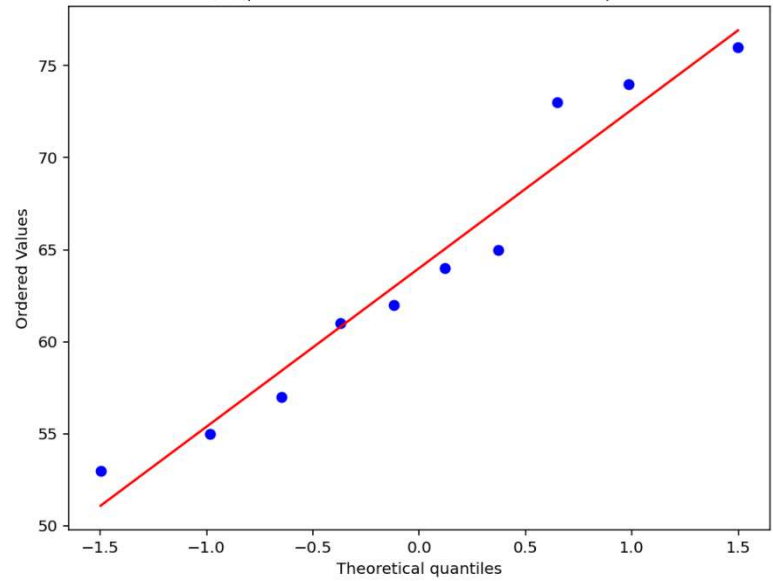
Data does not look like a normal distribution

Q-Q Visual Test for normality test – STEM RIGHT:

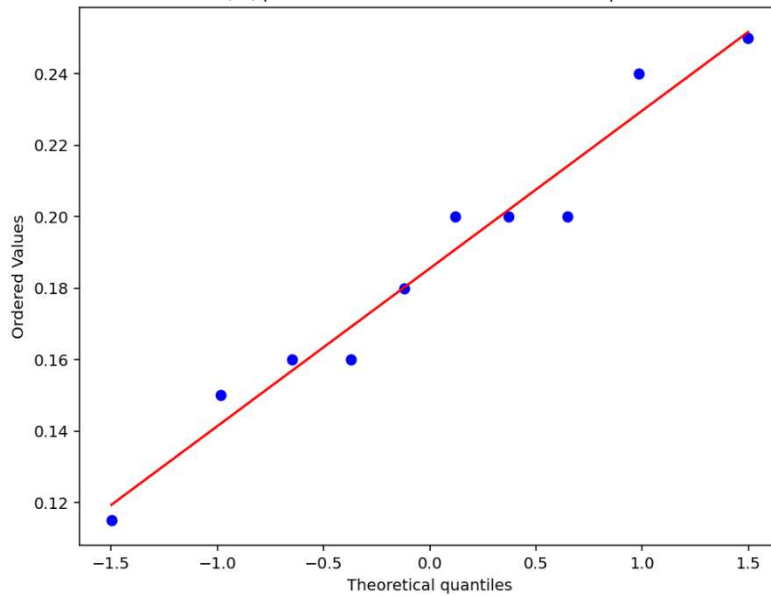
Q-Q plot for RIGHT STEM - GROUP1 - coppre



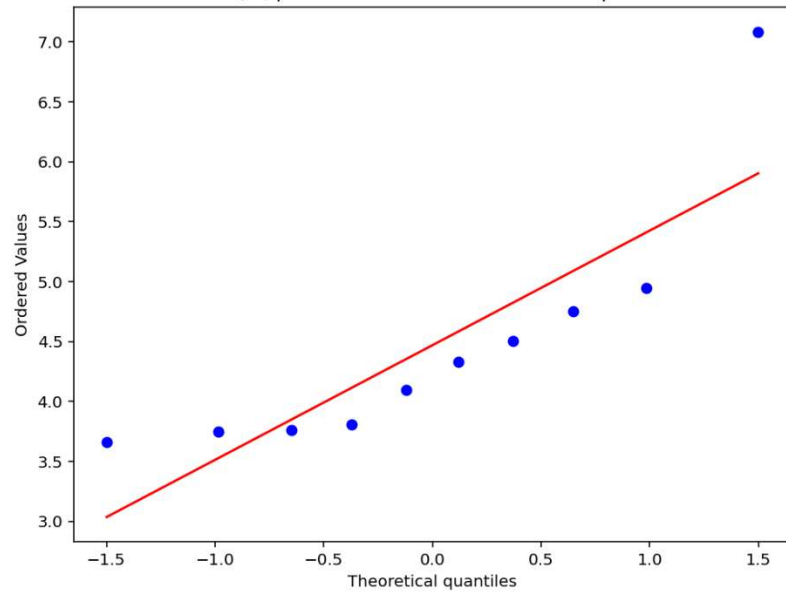
Q-Q plot for RIGHT STEM - GROUP1 - voxelpre



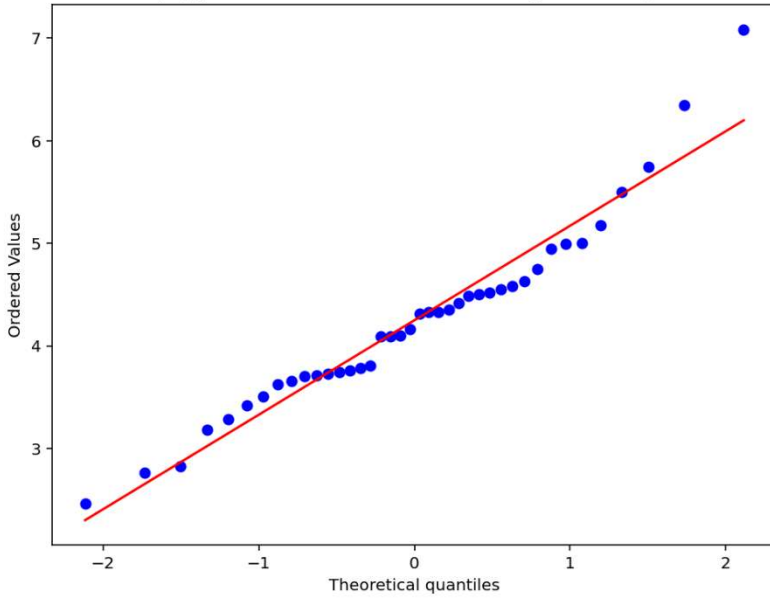
Q-Q plot for RIGHT STEM - GROUP1 - BSCpre



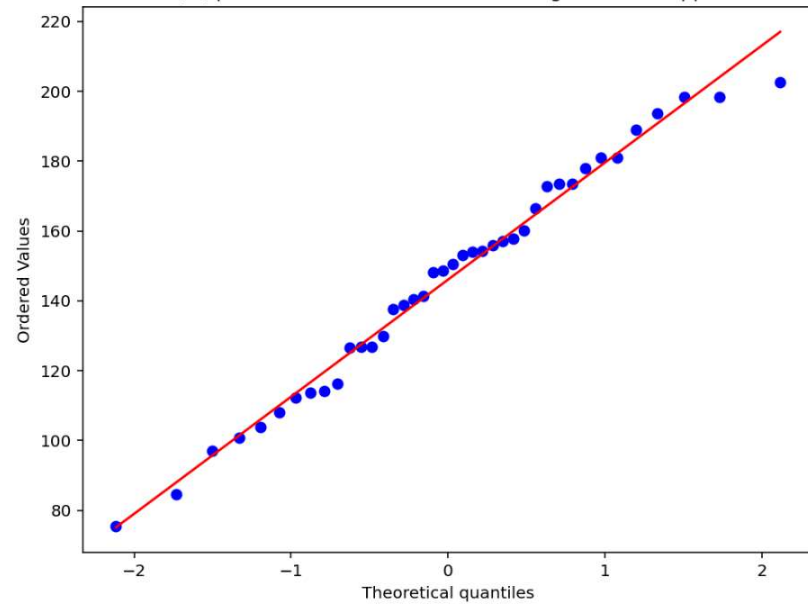
Q-Q plot for RIGHT STEM - GROUP1 - zpre



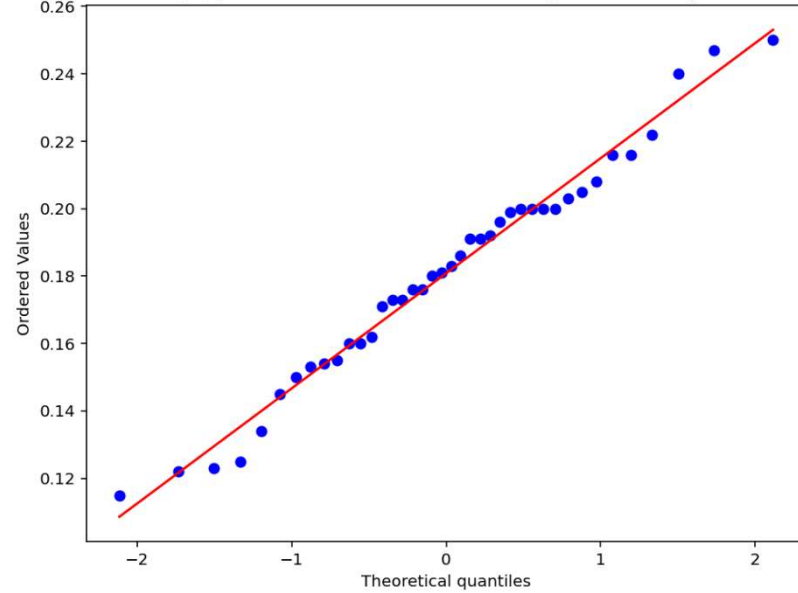
Q-Q plot for RIGHT STEM - GROUP1 Augmented - zpre



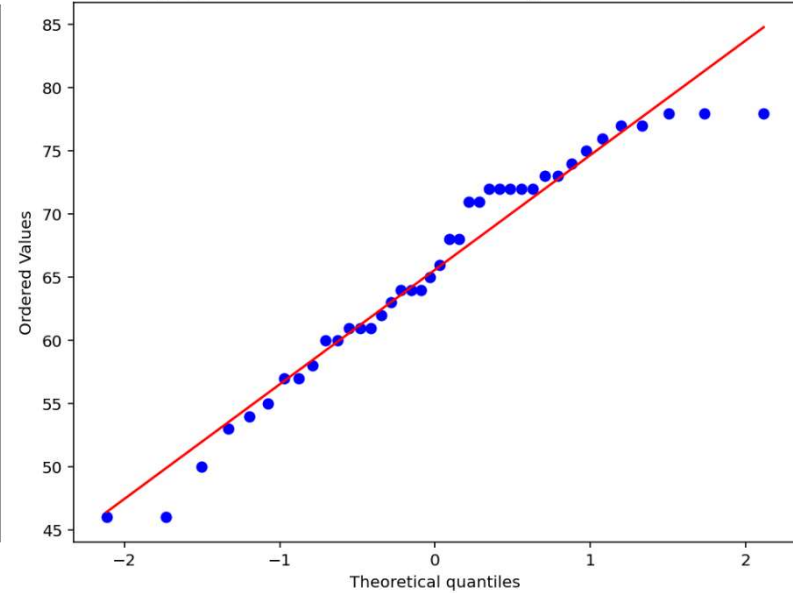
Q-Q plot for RIGHT STEM - GROUP1 Augmented - coppre



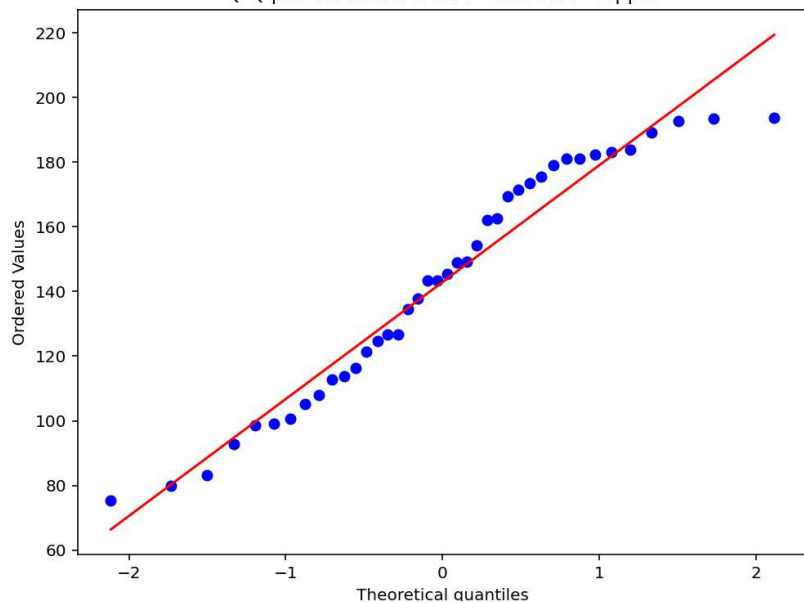
Q-Q plot for RIGHT STEM - GROUP1 Augmented - BSCpre



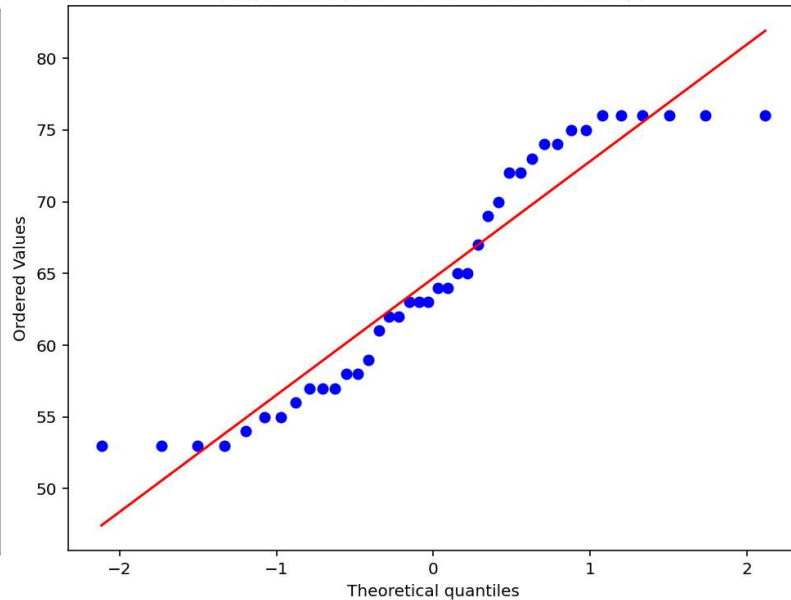
Q-Q plot for RIGHT STEM - GROUP1 Augmented - voxelpre



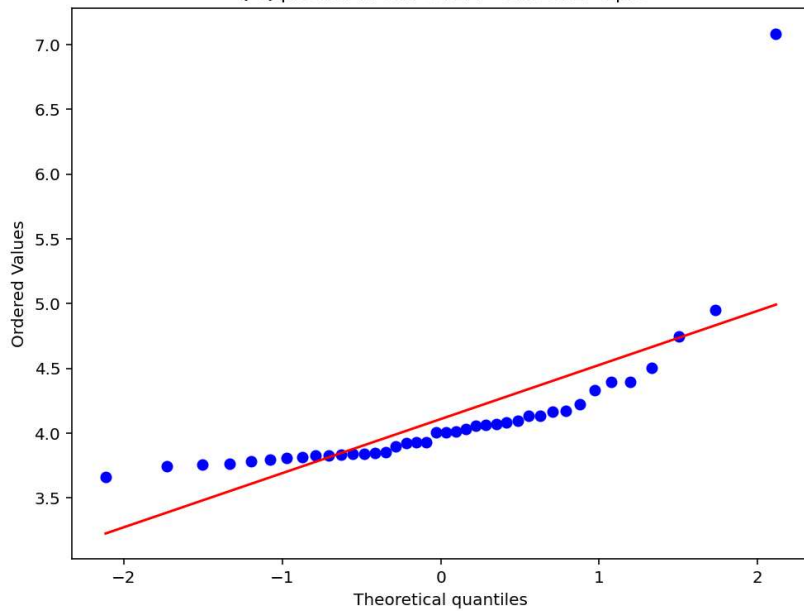
Q-Q plot for RIGHT STEM - GROUP2 - coppre



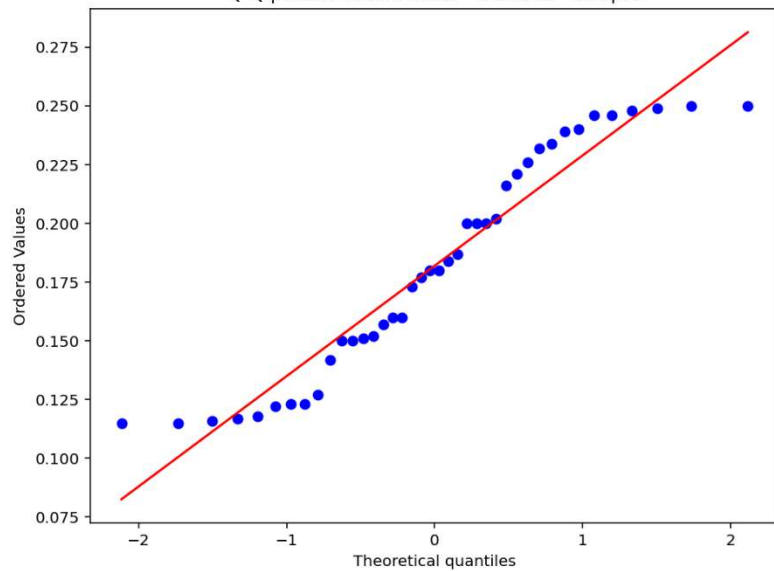
Q-Q plot for RIGHT STEM - GROUP2 - voxelpre



Q-Q plot for RIGHT STEM - GROUP2 - zpre

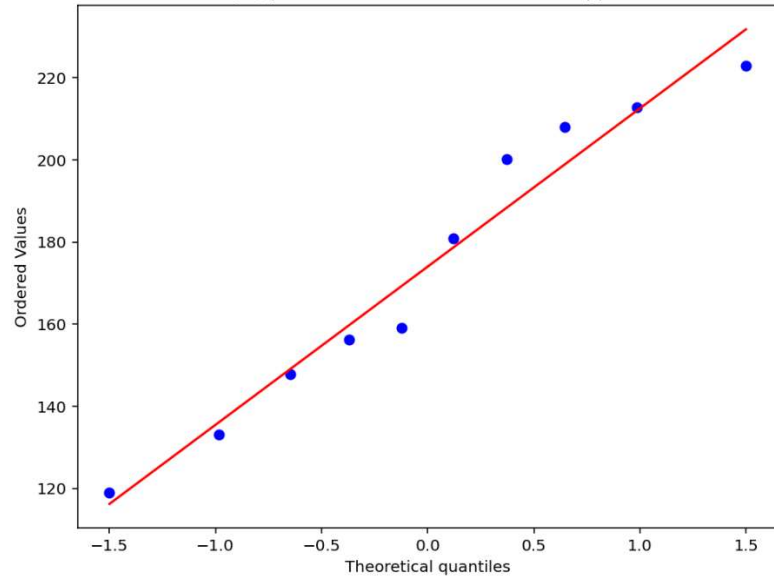


Q-Q plot for RIGHT STEM - GROUP2 - BSCpre

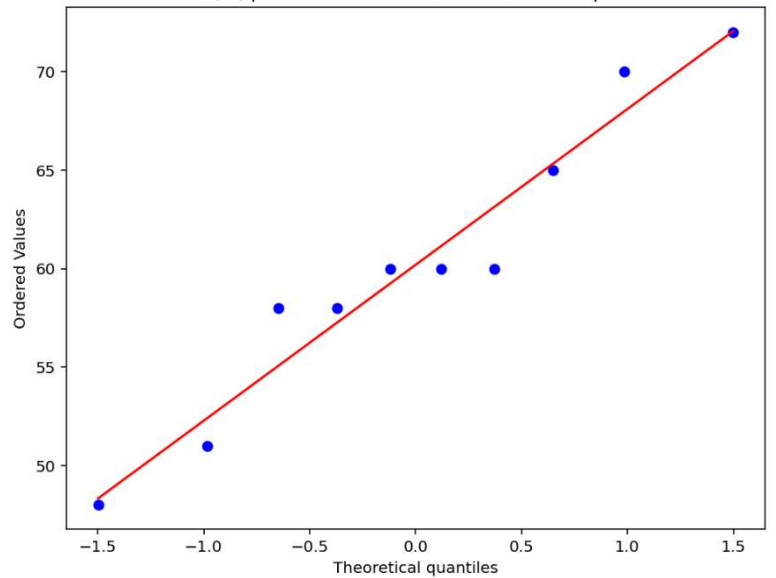


Q-Q Visual Test for normality test – STEM LEFT:

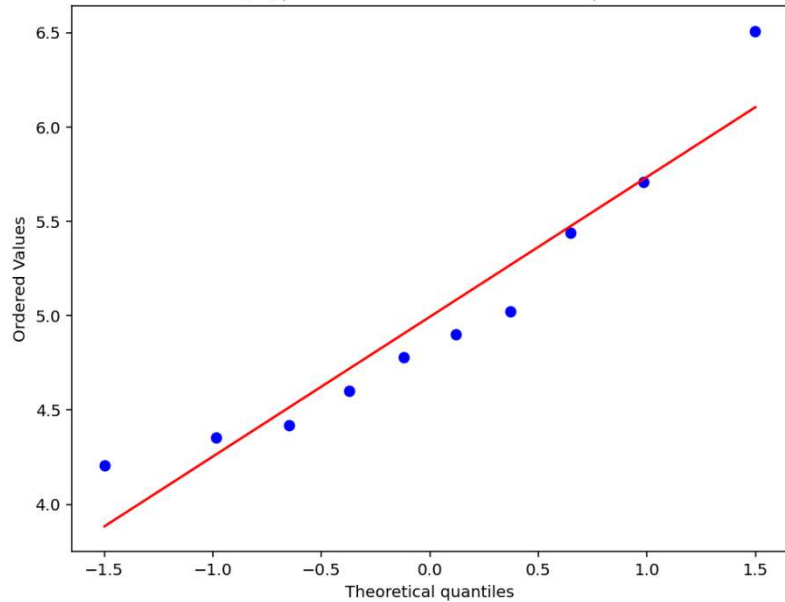
Q-Q plot for LEFT STEM - GROUP1 - coppre



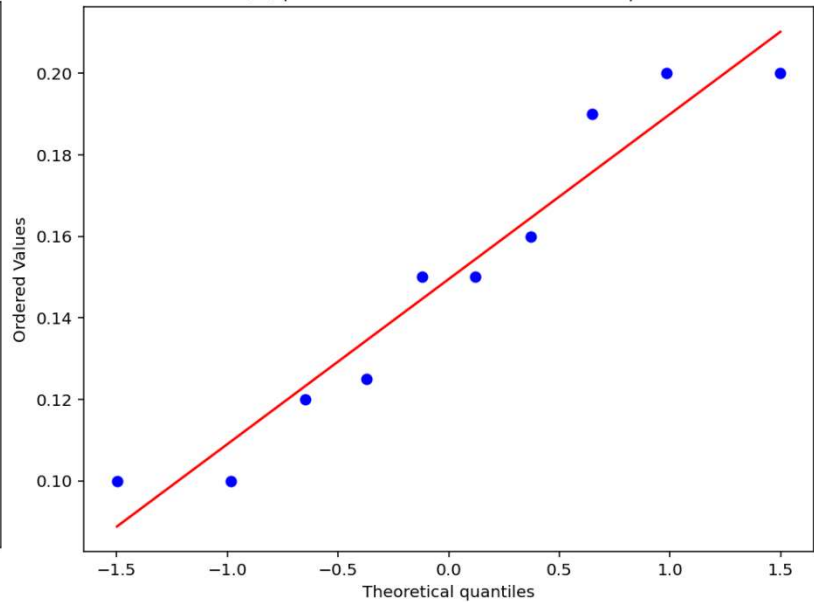
Q-Q plot for LEFT STEM - GROUP1 - voxelpre



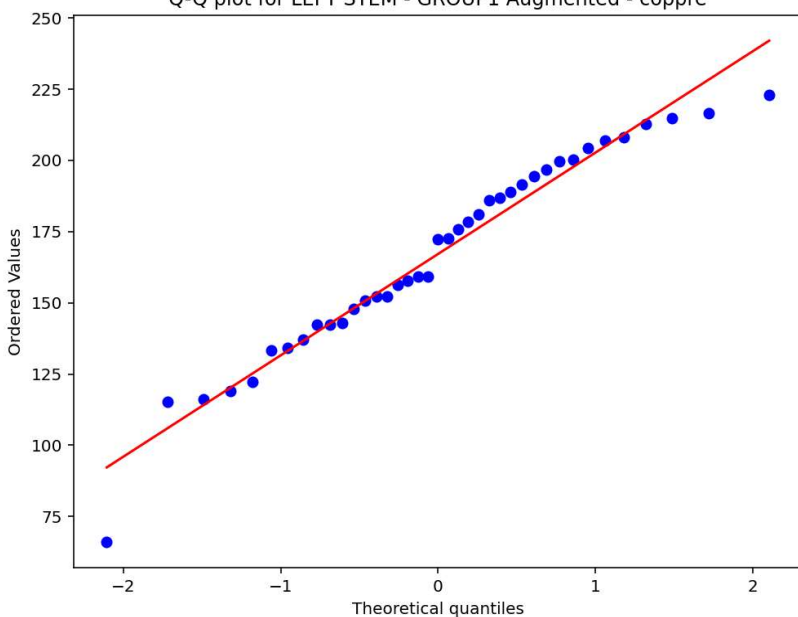
Q-Q plot for LEFT STEM - GROUP1 - zpre



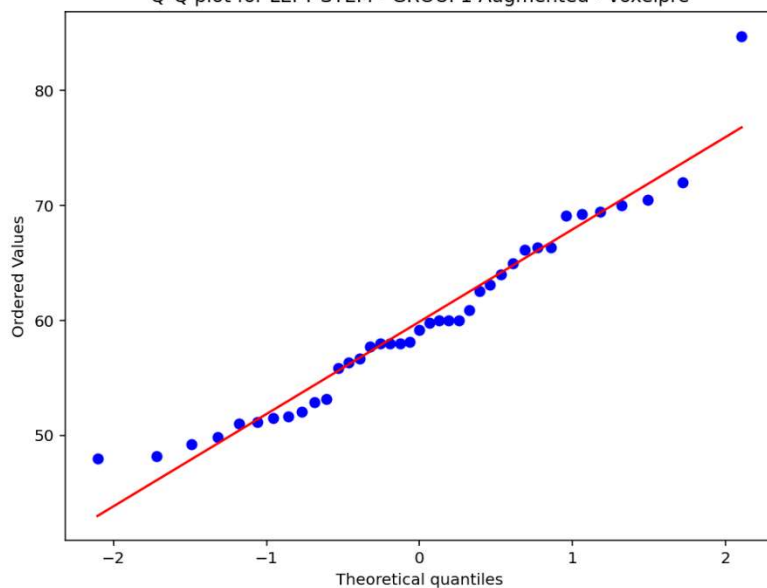
Q-Q plot for LEFT STEM - GROUP1 - BSCpre



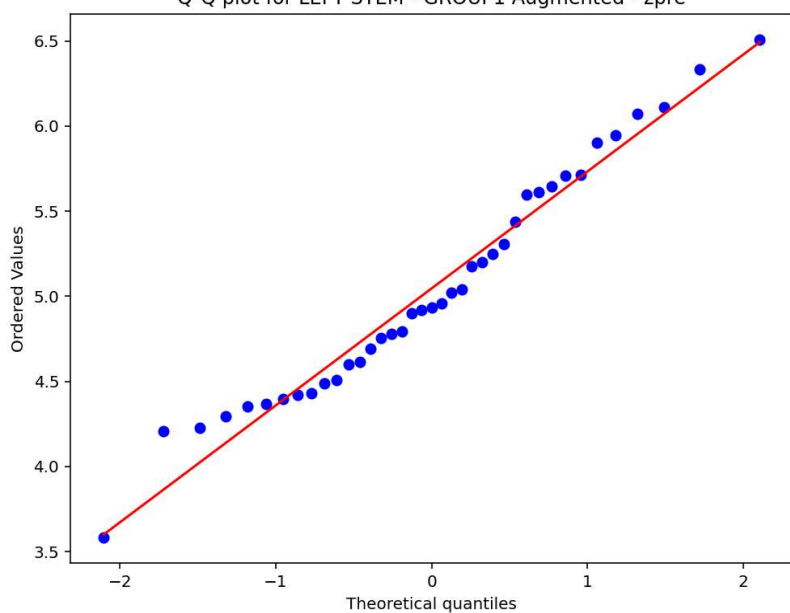
Q-Q plot for LEFT STEM - GROUP1 Augmented - coppre



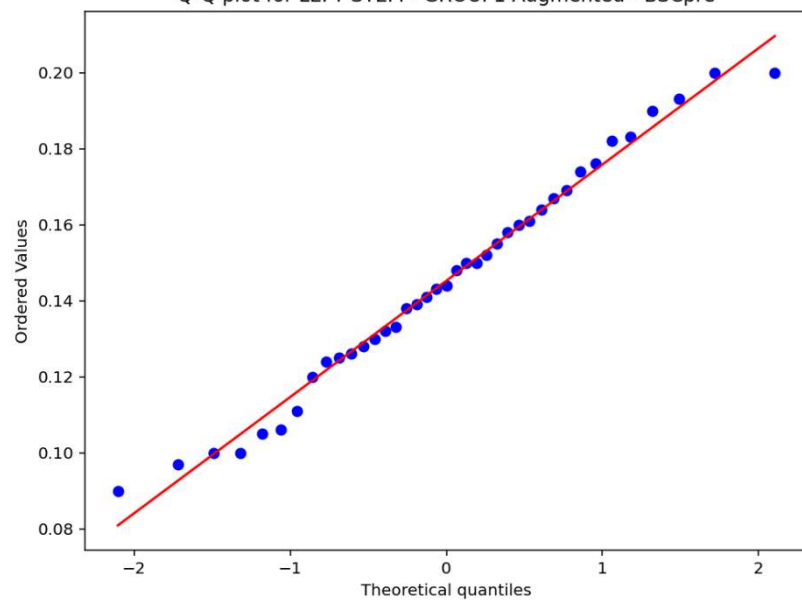
Q-Q plot for LEFT STEM - GROUP1 Augmented - voxelpre



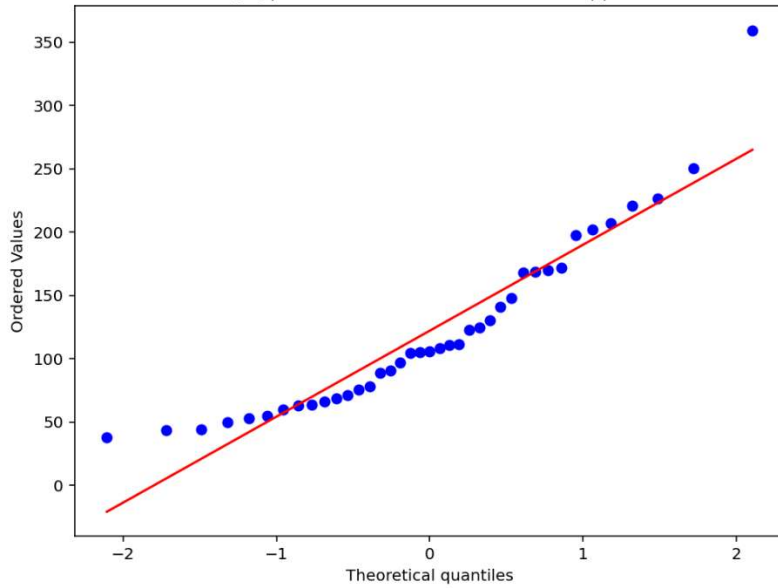
Q-Q plot for LEFT STEM - GROUP1 Augmented - zpre



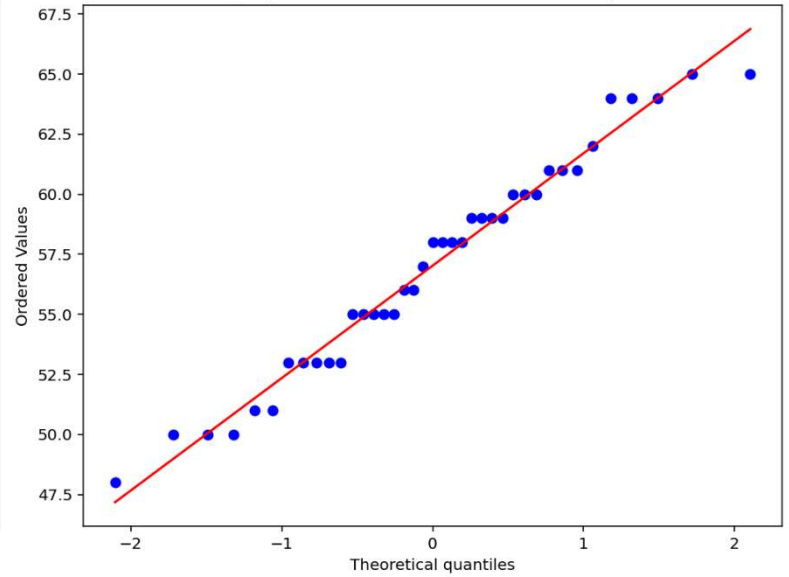
Q-Q plot for LEFT STEM - GROUP1 Augmented - BSCpre



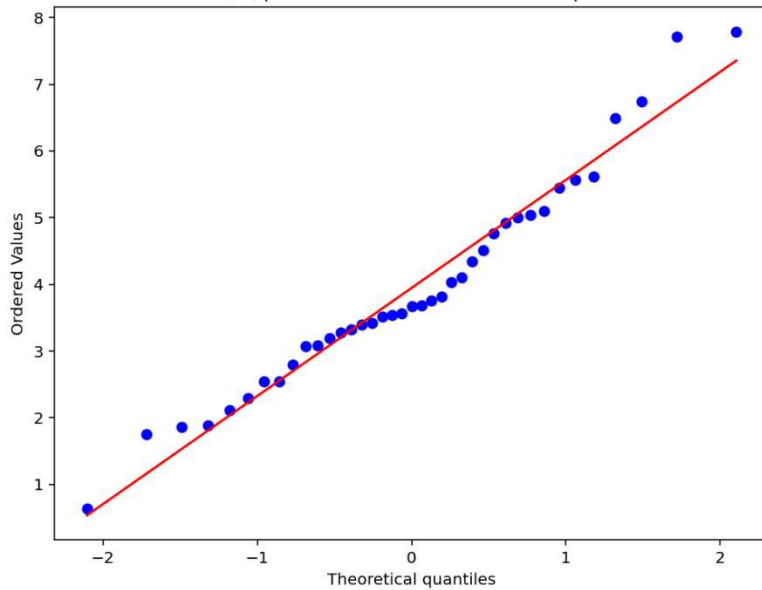
Q-Q plot for LEFT STEM - GROUP2 - coppre



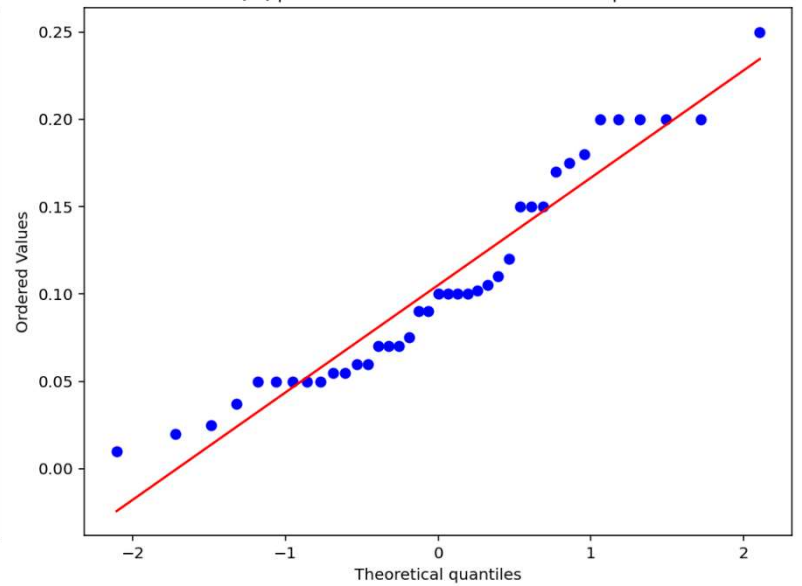
Q-Q plot for LEFT STEM - GROUP2 - voxelpre



Q-Q plot for LEFT STEM - GROUP2 - zpcre



Q-Q plot for LEFT STEM - GROUP2 - BSCpre



G1 vs G2 RIGHT ARI Assessment:

Adjusted Rand Index (ARI): -0.006749379652605459

G1-A vs G2 RIGHT ARI Assessment:

Adjusted Rand Index (ARI): 0.004353079563731805

G1 vs G2 LEFT ARI Assessment:

Adjusted Rand Index (ARI): 0.10715394968423232

G1-A vs G2 LEFT ARI Assessment:

Adjusted Rand Index (ARI): 0.3711814167322001